Report

Using the camera in ROS2

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1. Pre-requisites

- Ubuntu 24.04 Noble can run on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This is NOT a simulation. Gazebo is no longer used. The hardware is required.
- VSCode and its extension PlatformIO (PIO) should be installed.
- This package only introduces how to get started with the camera module 3.

2. Installing the camera on the RPI

The camera module v3 for RPI requires special attention for installation of the packages. If a USB camera is used, the process is much simpler. Using the camera module with the RPI is quite fragile. The latch to install the camera module can be tricky. Have a look at the following videos:
https://youtu.be/IAbpDRy-gc0?si=pMC79ZBYrfffU3NP
https://youtu.be/6poDg63DLVM?si=0s8pCWwj3Q08mxrH

Reference for installing the camera package:
https://github.com/christianrauch/camera_ros

Tasks	Code
Remote control the rpi via ssh	ssh rpi@ipaddr
Install dependencies	sudo apt update sudo apt install -y git build-essential meson ninja-build pkg-config libyaml-dev python3-pip sudo apt install -y libboost-dev libgnutls28-dev openssl libssl-dev libevent-dev sudo apt install -y qtbase5-dev libqt5core5a libqt5gui5 libqt5widgets5
Create workspace	mkdir -p ~/camera_ws/src cd ~/camera_ws/src
Install python3-colcon-meson	sudo apt -y install python3-colcon-meson
Get libcamera for camera module	git clone https://github.com/raspberrypi/libcamera.git
Get camera_ros	git clone https://github.com/christianrauch/camera_ros.git
Resolve binary dependencies and build workspace	source /opt/ros/jazzy/setup.bash cd ~/camera_ws/ rosdep install -y --from-paths src --ignore-src --rosdistro jazzy --skip-keys=libcamera colcon build --event-handlers=console_direct+

3. Testing the camera

Tasks	Code
Remote control the RPI via ssh	ssh rpi@ipaddr
On the RPI **Launch the camera node**	cd ~/camera_ws/ source /opt/ros/jazzy/setup.bash ros2 run camera_ros camera_node
On the PC **Launch rviz**	ros2 run rviz2 rviz2 add image camera/image_raw

Normally, in rviz, there is a topic called image, which can be selected to watch the stream from the camera. By default, the stream uses the highest resolution, which is likely yielding lagging video. To correct this, run the camera with lower resolution and listen to the *_compressed topic.

Tasks	Code
Remote control the rpi via ssh	ssh rpi@ipaddr
On the RPI **Launch the camera node**	cd ~/camera_ws/ source /opt/ros/jazzy/setup.bash ros2 run camera_ros camera_node --ros-args -p width:=320 -p height:=240 -p format:=YUYV
On the PC **Launch rviz**	ros2 run rviz2 rviz2 add image camera/image_raw_compressed

5. Future works

The robot is now complete. The camera feed is useful to navigate with the robot, however, much more intelligent features could be added with this camera, such as:

- Better localization with apriltag
- Computer vision: image recognition, object following, etc.